

Object Oriented Programming

1. What are object oriented concepts? What is difference between object-based, object-oriented and fully object-oriented language?
2. What is abstraction and encapsulation. Give real-life example.
3. What is polymorphism? What are its types? Explain them with examples.
4. What is object slicing? Explain object slicing in context of up-casting?
5. What is down-casting and when it is required? Explain with code.
6. What do you know about association, composition and aggregation? Explain with the help of example.
7. What are different types of inheritance? Explain with the help of example. What are problems with multiple inheritance?
8. What is difference between interface, abstract class and non-abstract class? Which one to use where?
9. Which are the different types of design pattern? Explain singleton design pattern.

Java Programming

1. What is a collection framework? Explain any 2 classes in Java Collections.
2. What is functional programming? Explain stream programming with examples.
3. start() vs run() in threads. What happens if you call run() instead of start() for an object of class that extends Thread?
4. What is the use of Garbage Collector in Java language?
5. Can you override the static method?
6. If multiple interfaces have the same method declaration, does it cause name collision or any other problem when they are implemented in a class?
7. Explain default methods in the interfaces.
8. What is the initial capacity of a collection and what happens when it is exhausted? Explain with examples.
9. Is HashMap thread safe? How to make it thread safe?
10. What is the difference between == and equals() in Java? Explain in context of primitive values and objects with examples.
11. What is the lambda expression in Java? How data types of lambda arguments and return types are determined?
12. What is a functional interface? Which functional interfaces are predefined and where they are used?